Part II — Post-hoc validity of $\mathcal{C}$ at $\hat{\theta}(\mathcal{D})$

research/formal (working)

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Depends on `common_setup.tex` *and Part I. Goal (iii): data-randomize selection;* ***never*** *recalibrate $\mathcal{C}$. Set-valued transfer:* `proofs/sol_gaps/2026-07-27_setvalued_thm2.txt`.

1 Targets

Definition 1 (Failure indicator)**.** *For a family $C^{(q)}(X; D, \theta)$ calibrated at miscoverage level q,*

$$\ell_q(W, \theta) := \mathbf{1}\{Y \notin C^{(q)}(X; D, \theta)\}, \quad W = (D, X, Y).$$

Writeup (1) is the special case $q = \alpha$ for prespecified *θ.*

2 Goal (i) — set-valued stability transfer

Assumption 1 (Joint fixed-θ validity on the selection support)**.** *Let $\Theta_{\mathrm{val}} \subseteq \Theta$ be the (problem-specific) set of hyperparameters for which writeup (1) holds at miscoverage level q. Assume*

$$\sup_{\theta \in \Theta_{\mathrm{val}}} \mathbb{P}\big\{Y \notin C^{(q)}(X; D, \theta)\big\} \le q. \tag{FV}$$

Probability is over the joint experiment (D, X, Y) (and set randomization); $\theta \in \Theta_{\mathrm{val}}$ is held fixed. Notation: $C^{(q)}(X; D, \theta)$ is writeup $\mathcal{C}(X; \mathcal{D}, \theta)$ when the classical miscoverage level in (1) equals q (superscript = level; third argument = hyperparameter θ). Do not read the superscript as replacing θ. Score-threshold families: $\Theta_{\mathrm{val}} = \{\theta : F(\theta) \ge 1 - q\}$; do not claim (FV) on all of $[0, 1]$.

Theorem 1 (Set-valued stability transfer)**.** *Let $\hat{\theta} = A(D)$ be (η, τ, ν)-stable with oracle $A_0 \perp W$ and good event G of probability $\ge 1 - \nu$ such that on G, $K_w(B) \le e^{\eta} Q_0(B) + \tau$ for all measurable $B \subseteq \Theta$ (Zrnic–Jordan Def. 2 / (ST)). Assume oracle validity $\mathbb{P}\{Y \notin C^{(q)}(X; D, A_0)\} \le q$ (implied by (FV) if $A_0 \perp W$ and A_0 is supported on Θ_{val};* ***(OV) alone suffices*** *— (FV) is only a convenient sufficient condition). Assume C's downstream randomization is a common Markov kernel given (W, θ) (no unaccounted coupling of A into the construction of C beyond plugging θ). Then*

$$\mathbb{P}\big\{Y \notin C^{(q)}(X; D, \hat{\theta})\big\} \le e^{\eta} q + \tau + \nu.$$

In particular, with $q = \delta e^{-\eta}$,

$$\mathbb{P}\big\{Y \notin C^{(\delta e^{-\eta})}(X; D, \hat{\theta})\big\} \le \delta + \tau + \nu.$$

Remark 1 (Proof idea)**.** *Failure section $B_q(w) = \{\theta : y \notin C^{(q)}(x; d, \theta)\}$; apply (ST) to the data-dependent section; add ν-mass of G^c; use $A_0 \perp W$.*

Remark 2 (Goals (i)–(ii) vs (iii))**.** *Using $C^{(\delta e^{-\eta})}$ instead of $C^{(\alpha)}$ is classical-level inflation (recalibration of the* level*). That addresses (i)–(ii) when η is fixed. Goal (iii) forbids rebuilding $\mathcal{C}$; it drives $\eta, \tau, \nu \to 0$ by randomizing* selection *only.*

3 Goal (ii) — inflation

Definition 2 (Inflation)**.** $\mathrm{Infl} := \mathbb{P}\{Y \notin C^{(\alpha)}(X; D, \hat{\theta})\} - \alpha$. *Theorem 1 implies* $\mathrm{Infl} \le (e^{\eta} - 1)\alpha + \tau + \nu$ *when* $q = \alpha$ *is used without level change.*

4 Goal (iii) — vanishing stability correction

Corollary 1 (Vanishing inflation at fixed classical level — W5/(iii))**.** *Let* $\{A_r\}$ *be* (η_r, τ_r, ν_r)*-stable with* $\eta_r \to 0$ *and* $\tau_r + \nu_r \to 0$*, and suppose randomizing* A_r *does* ***not*** *alter the fixed-*θ *map* $(D, X) \mapsto C^{(\alpha)}(X; D, \theta)$ *(same classical level* α *as in writeup (1);* no *level change). If (OV) holds for the oracle of* A_r *at level* $q = \alpha$ *and the fixed-*θ *map is unchanged, then Theorem 1 with* $q = \alpha$ *yields*

$$\mathbb{P}\big\{Y \notin C^{(\alpha)}(X; D, A_r(D))\big\} \le e^{\eta_r}\alpha + \tau_r + \nu_r,$$

hence

$$\mathrm{Infl}_r \le (e^{\eta_r} - 1)\alpha + \tau_r + \nu_r \to 0.$$

In the limit $\eta = \tau = \nu = 0$*, classical (1) holds at the selected parameter with the* same $C^{(\alpha)}$*. (The optional path* $C^{(\delta e^{-\eta})}$ *belongs only under goals (i)–(ii); it is* ***not*** *the (iii) mechanism.)*

Definition 3 (Randomized selection map)**.** $(\mathcal{D}, \xi) \mapsto \tilde{\theta}(\mathcal{D}; \xi)$ *with* ξ *independent of the test point; fixed-*θ *map of* $\mathcal{C}$ *unchanged.*

Corollary 2 (Forbidden repairs)**.** *Sample-splitting re-estimation of* $\mathcal{C}$ *and threshold/score recalibration of* $\mathcal{C}$ *are out of scope as answers to (iii) (contrast literature only).*

Problem 1 (Remaining)**.** *Instantiate* A_r *via* `part_i_randomized_design.tex` *(RNM / Lap* $b = 2\varepsilon/\eta_r$*, or soft-argmin* $\beta = \eta_r/(2\varepsilon)$*) with* $\eta_r \to 0$*; verify (FV) and Ass. concentration* $\varepsilon(n), \nu(n)$ *for the project's* $(\hat{f}, \hat{g})$ *and* $\mathcal{C}$*.*